



Robots for economic development

Massimiliano Calì ^{a,1}, Giorgio Presidente ^{b,*}

^a World Bank, United States of America

^b Bocconi University, Italy

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ABSTRACT

Recent evidence suggests that in manufacturing, automation technologies entail a trade-off between productivity gains and employment losses for the economies that adopt them. This paper casts doubts on such trade-off in the context of a developing country. It shows significant employment gains from automation in Indonesian manufacturing during the years 2008–2015, a period of rapid increase in robot imports. Analysis based on manufacturing plant data provides evidence that the absence of this trade-off is due to diminishing productivity returns to robot adoption. As a result, the benefits from automation could be particularly large for countries at early stages of industrialization.

1. Introduction

Ever-more ubiquitous automation technologies present a seeming trade-off for the economies that adopt them. On one hand, they generate significant productivity gains for adopting firms. On the other hand, the productivity gains are associated with the displacement of some workers and of non-adopting competitors (Acemoglu et al., 2020b). These channels typically result in a net negative employment effect in manufacturing, which is the sector responsible for the bulk of automation. This negative employment impact on local manufacturing has been documented for advanced and industrialized emerging economies. These include — among others — the United States (Acemoglu and Restrepo, 2020), Germany (Dauth et al., 2019), the United Kingdom (Chen et al., 2021), Spain Koch et al. (2021) China (Giuntella and Wang, 2019) and Mexico (Artuc et al., 2019).²

Much less is known about the impact of automation in countries at earlier stages of industrialization, which include most developing countries today. This is an important gap, as the penetration of automation

in developing countries is expected to grow significantly over the next decades (Hallward-Driemeier and Nayyar, 2017), especially in Asia. For instance, Fig. 1 shows that after 2015 — the last year of our sample — the growth rate of robot imports exceeded fifty percent in several large Asian developing economies.

This paper makes two contributions. The first is to document that, in contrast to the evidence based on industrialized economies, the aggregate local employment impact of industrial robots in manufacturing is positive in a developing country. To do so we focus on the case of Indonesia in the late 2000s/early 2010s, when the adoption of industrial robots, an important class of automation technologies, grew rapidly. In particular, our results cast doubts to the idea that the adoption of capital-intensive technology like robots can explain the poor employment performance of manufacturing in countries at early stages of industrialization (Diao et al., 2021).

The second contribution is to shed light on the microeconomic mechanism underlying the positive impact of robots we document

* Corresponding author.

E-mail address: giorgio.presidente@unibocconi.it (G. Presidente).

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² Dauth et al. (2019) find evidence that the adoption of robots has led to positive employment spillovers on *non-manufacturing* industries, likely due to complementarities between the expanded production following robot adoption in manufacturing and manufacturing services. Instead, the focus on this paper is on manufacturing.

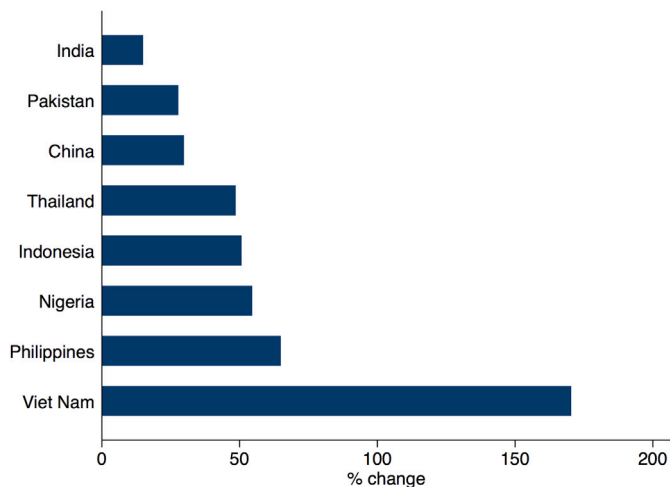


Fig. 1. Growth in machinery and robot imports: 2015–2020.

Notes: this figure shows the percentage change in import values of machinery and industrial robots (HS 849750).

Sources: Comtrade.

at the aggregate local labor market-level. We argue that automation technologies are subject to strongly diminishing productivity returns. Therefore, plants with untapped automation possibilities, which are typically abundant in developing countries, experience the largest productivity gains from automation. We provide evidence that at low-initial levels of industrialization, these gains are so large that they offset the labor displacement effect, thereby resulting in aggregate net employment gains. This contrasts with plants in advanced economies, where producers have exploited most available productivity-enhancing opportunities and the marginal impact of automation on productivity is too limited to compensate the labor displacement effect.

The seminal study of [Graetz and Michaels \(2018\)](#) has provided some preliminary evidence of diminishing productivity returns to automation. We build on that intuition, testing it using micro data, and use the findings to rationalize differences in employment impacts at the aggregate local-level. Our results also speak to the intuition in [Acemoglu and Restrepo \(2019\)](#) that the largest displacement effects of “so-so automation technologies” — i.e. non disruptive technologies — are more likely to take place at later stages of industrialization. In those instances, humans have a strong comparative advantage in performing complex tasks that are not yet automated and so automation results mainly in cost reduction, rather than in substantial productivity gains.

[Fig. 2](#) provides some suggestive evidence in support of these ideas. It shows that the correlation between robot penetration and employment is negative for OECD countries — which have higher robot penetration — and positive for non-OECD countries. In the paper, we formalize this intuition and examine it empirically along two levels of analysis. First, to identify the employment effect at the local labor market-level, we follow the standard approach of constructing a Bartik-style measure ([Bartik, 1991](#)) interacting baseline industry shares in a local labor market with industry-specific robot imports (e.g. [Acemoglu and Restrepo, 2020](#); [Dauth et al., 2019](#)). We address the endogeneity of robot imports by instrumenting it with a measure of *automation possibilities*, based on industry-specific robot penetration in countries ahead of Indonesia in terms of robot adoption. A large battery of tests provides confidence on the robustness of the positive employment effect of robots in Indonesia.

We then turn to plant-level data to shed light on the mechanisms behind the labor market-level results. The main analytical challenge is that we do not observe robot adoption at the plant-level. To overcome this issue, we follow ([Autor et al., 2003](#)) and assume that industrial robots are best suited to performing routine manual tasks. We document that most workers performing routine task-intensive occupations

in Indonesian manufacturing have secondary school as their educational attainments, which we directly observe for each plant in our data.

We quantify the exposure of plants to robots by interacting yearly imports of robots by 3-digit industry with plants’ share of secondary education workers in the base year. We perform a battery of tests to assess the validity of our exposure measure, and show that it is strongly correlated with plant-level investment in machinery and equipment, which includes industrial robots. We also test the robustness of our results against other changes in economic activity at the industry level—including foreign direct investment (FDI)—as well as other forms of technical change that might be correlated to robot adoption, such as skill-biased and routine-biased technical change, which has been shown to impact even occupations involving non-manual routine tasks (e.g. [David and Dorn, 2013](#)). Placebo tests based on an exposure variable interacting the share of secondary education workers with other forms of non-robot capital, suggest that our findings are not driven by investment in general or the building of new plants.

Consistent with the predictions of a task-based model (e.g. [Acemoglu and Restrepo, 2018](#)), the micro-level evidence supports the hypothesis of diminishing employment and productivity returns to robot adoption. We find that plants in the bottom tercile of initial robot penetration benefit the most from adoption, with returns to adoption more than double the average returns. Conversely the gains are close to zero for plants in the higher terciles of initial penetration.

Indonesia is a suitable laboratory for our analysis. During the period of analysis the penetration of robots was considerably lower than in the other countries with available evidence. The number of robots in the country was very limited before the beginning of our sample in 2008 and accelerated swiftly thereafter. This acceleration was highly heterogeneous across sectors. By the end of the sample period in 2015, the penetration of robots in the most automated industries, such as Motor Vehicles, was similar to that of advanced economies. Other industries, such as Textiles, have seen no penetration throughout the period. Indonesia has also a rich set of micro- and local-level data on manufacturing plants and employment, enabling us to employ methods comparable to other studies to identify the effects of automation and explore the underlying mechanisms.

The paper is related to the large literature on firm performance in developing countries (see [Verhoogen, 2020](#) for a review). It focuses in particular on automation technologies. An emerging literature studies the impact of automation technologies in advanced and emerging economies, typically showing negative aggregate effects on employment and positive on firms’ performance (e.g. [Acemoglu et al., 2020b](#); [Koch et al., 2021](#); [Acemoglu and Restrepo, 2020](#); [Dauth et al., 2019](#); [Graetz and Michaels, 2018](#); [Acemoglu et al., 2020a](#); [Aghion et al., 2020](#); [Humlum, 2019](#); [Bessen et al., 2020](#); [Dixon et al., 2020](#); [Bonfiglioli et al., 2020](#); [De Vries et al., 2020](#)). [Dauth et al. \(2019\)](#) represent a partial exception to these findings as they find that the displacement impact of automation on manufacturing jobs in Germany is offset by new jobs in services. By showing net employment creation effects of automation in manufacturing itself, our results depart from those based on advanced economies, including ([Dauth et al., 2019](#)).

We argue that the key factor accounting for the difference in results is our focus on a developing country at the early stage of industrial automation, along with diminishing returns to robot adoption. That sets our findings apart from other studies on developing countries at more advanced stages of automation, including [Giuntella and Wang \(2019\)](#) and [Artuc et al. \(2019\)](#), which find negative employment impact of automation. Differently from these studies, who focus on local labor markets, we also add micro-level evidence at the plant level to document the underlying mechanism.

This paper is also related to the literature examining the impact of automation in advanced economies on developing countries through international trade linkages and reshoring ([Kugler et al., 2020](#); [Artuc](#)

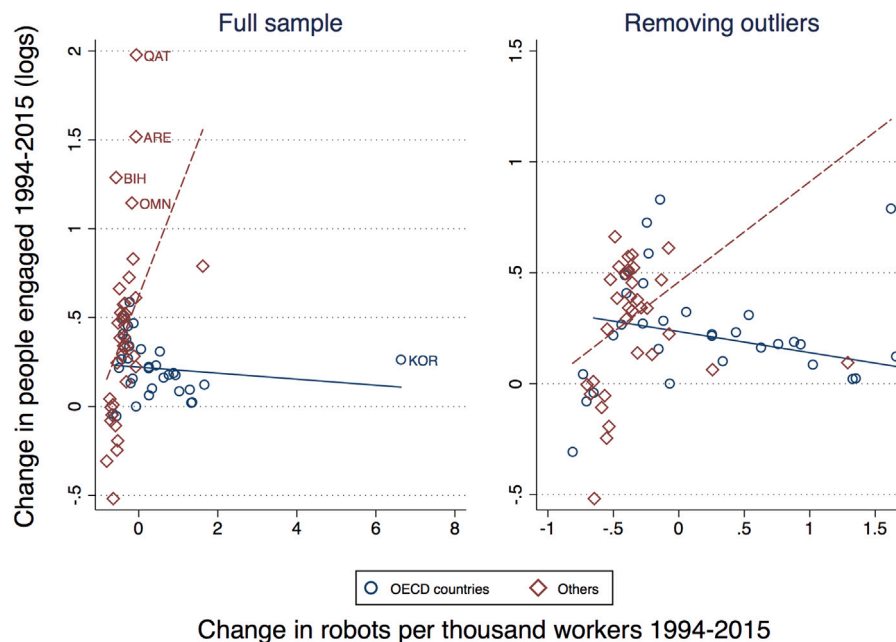


Fig. 2. Robot penetration and employment changes: OECD vs. non-OECD countries.

Notes: this figure plots the correlation between the change in residuals from a regression of log-employment on the share of population above 55 years old over population between 20 and 49 years old, and changes in robot penetration over the same period. Robot penetration is defined as the stock of industrial robots per thousand employed workers.

Sources: IFR; PWT; World Bank.

et al., 2019; Faber, 2020; Artuc et al., 2018). In contrast to that literature, this paper provides novel evidence on the impact of automation in a developing country on its own economy.

The rest of the paper is organized as follows: Section 2 presents the positive long-run impact of robots on local labor markets in Indonesia. Section 3 examines candidate mechanisms underlying the aggregate impact, decreasing returns to automation; Section 4 explores decreasing returns to automation using plant-level data; and Section 5 concludes.

2. The positive employment impact of robots on local labor markets

This section provides evidence of the labor market impact of industrial robots in Indonesia, a developing country at the early stage of adoption of automation technologies, during a period of rapid increase in robot imports.

Indonesia is one of the largest middle-income countries (MICs). It has been historically relatively closed, with a trade-GDP ratio well below the average of MICs during the period of our analysis (48.6% versus the average 52.4% for MICs and 61.7% for another large middle-income economy as Mexico).³ Its manufacturing sector represents a sizable share of its economy. During the 2008–15 period it accounted for almost 23% of GDP, above the average for middle-income countries (21%).

However, Indonesia's manufacturing was less productive relatively to comparable countries, with a per capita real value added 23% below that of the average MIC in 2008–15, and around three times smaller than China and Mexico. Such difference is amplified for manufacturing export per capita, the value of which for Indonesia during 2008–15 was only 37.9% that of the average MIC, 19.4% and 11.5% those of China and Mexico respectively. Part of this gap was due to Indonesia's relatively unsophisticated manufacturing exports. During the period of analysis, only 29.7% of Indonesia's manufacturing exports were classified as medium or high-tech, against 58.8% for China and 78.5%

for Mexico. These differences are also consistent with the lower level of adoption of robots in Indonesia, relatively to that in countries such as China and Mexico.

2.1. Robots and local labor market data

We match imports of industrial robots from the International Federation of Robotics (IFR) and Indonesian labor force survey data on employment in local labor markets (Survei Tenaga Kerja Nasional—Sakernas).

Industrial robots are defined by the International Standards Organization as *an automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications*.⁴ The autonomy and physical dexterity typical of robots make them particularly suitable to perform tasks otherwise performed by human workers.

The IFR collects data from each national robotics association. Since almost all robot suppliers are members of national associations, the dataset includes virtually all robots used worldwide. An advantage of the data is that the IFR has a common protocol to count robots, so that it ensures consistency across countries and years. For each country-year pair, information is available for manufacturing and non-manufacturing industries.

The IFR provides a relatively fine industry breakdown for manufacturing, which comprises 13 industries and it is compatible with 3-digit ISIC classification. On the other hand, non-manufacturing only includes 6 industries, reflecting the generally low level of robot adoption outside manufacturing. During our period of analysis, manufacturing industries account for 99.9% of robot installations in Indonesia.⁵ Given this fact and that the bulk of our analysis is based on manufacturing plants, we only consider robot adoption in the 13 IFR manufacturing industries.

⁴ ISO 8373:2012.

⁵ This share is comparable to that in advanced economies, according to IFR data. For instance (Acemoglu et al., 2020b) report that in the United States, roughly 1% of robots are in non-manufacturing industries.

³ The data in this sub-section is derived from the World Bank's World Development Indicators.

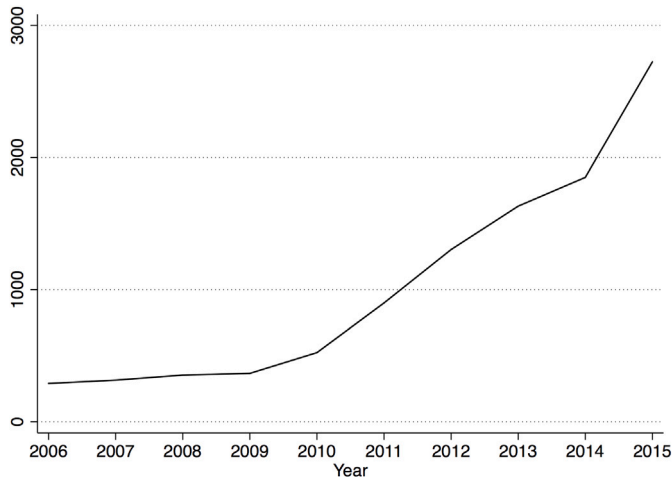


Fig. 3. Total number of robots used in the Indonesian manufacturing sector.
Notes: this figure shows the total number of industrial robots used in the manufacturing sector over the years of the sample.
Source: IFR.

Details about the industry-level data on robots are presented in Online Appendix C.1.

The data shows that the total number of robots shipped to the Indonesian manufacturing sector was low and roughly constant up to 2009, when the country had a stock of approximately 350 units in use (Fig. 3).⁶ Since 2010, the pace of increase in robots import experienced a significant uptick with the total stock increasing almost 10-fold by 2015. Our analysis exploits this large jump in adoption of robots to identify their impact.

The building block of the empirical methodology is industry-level robot penetration, defined as the number of industrial robots shipped to industry i (measured at 3-digit ISIC code) in year t , divided by the number of workers in industry i (in thousands):

$$\frac{R_{i,t}}{L_{i,0}}$$

To relieve potential endogeneity concerns, we fix the number of industry workers (the denominator) to the base year value, in the same vein as (Acemoglu and Restrepo, 2020).

The aggregate trend presented in Fig. 3 hides a large amount of industry heterogeneity in the use of robots. Fig. 4 shows that the number of robots per thousand workers used in manufacturing is much higher in Motor vehicles and Rubber and plastics than in other industries.⁷

While the absolute number of industrial robots in Indonesia is low compared to advanced economies, the penetration of robots per worker in the most automated industries is comparable to the average penetration in OECD countries.⁸ Therefore, it is plausible to expect that industrial robots can have a meaningful impact in Indonesia, comparable to that in advanced economies, at least for such industries.

⁶ Indonesia does not produce robots domestically. Therefore, IFR data coincide with total imports of robots from Comtrade. However, unlike IFR data, Comtrade does not provide an industry breakdown.

⁷ While Motor vehicles is by far the most automated industry worldwide, the high concentration of robots in Rubber and plastics is more peculiar to Indonesia. This is one of the largest manufacturing industries in Indonesia particularly for auto-parts, such as tires, which often employ state-of-the-art technologies of production.

⁸ For instance, Acemoglu and Restrepo (2022) reports that the number of industrial robots per thousand industrial workers in the United States was 8.4 in 2014.

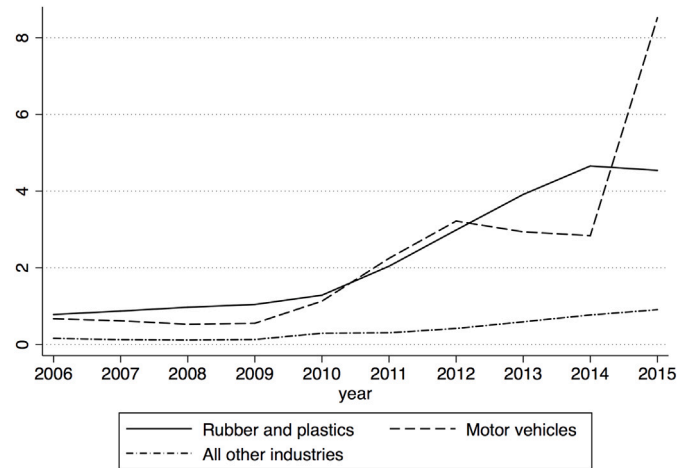


Fig. 4. Penetration of robots in the Indonesian manufacturing sector, by industry.
Notes: this figure shows the number of industrial robots per thousand workers used in selected industries over the years of the sample.
Source: IFR, SI.

Labor force survey data comes from the Indonesia National Statistics Bureau (BPS), which collects data on a large cross-section of workers bi-annually.⁹ Regencies represent the second level of sub-national administrative divisions in Indonesia, the first being provinces, which are composed of multiple regencies. A number of features make them a reasonable proxy for local labor markets in Indonesia. First, the mobility of labor is limited across regencies. In 2010 for example, Sakernas data shows that only 5 percent of the workforce worked in a different regency than their residence.¹⁰ Second, following Indonesia's 1999 decentralization reforms, regencies hold significant administrative powers, including in the labor markets, such as the minimum wage setting. We use the pre-decentralization reform division in 292 regencies to ensure the consistency of the analysis over time.

The details of the labor market data are provided in Section 2.1. Table A.1 presents the summary statistics.

2.2. Local labor market analysis: Empirical approach

To identify the effects of robots on local employment, we follow the standard approach of constructing a Bartik-style measure (Bartik, 1991) interacting baseline industry shares in a local labor market with industry-specific robot imports (Acemoglu and Restrepo, 2020; Dauth et al., 2019; Artuc et al., 2019; Giuntella and Wang, 2019).

We set 2007 as the base year as Sakernas data is representative at the regency-level only since 2007. For each industry in the base year, we compute industry i share of employment in regency r total employment:

$$s_{i,r} = \frac{L_{r,i,2007}}{L_{r,2007}} \quad (1)$$

we then use these shares to aggregate industry-specific changes in robot penetration weighted by the initial regency's industry composition. Accordingly, we define Indonesian regency-level exposure to changes in robot penetration as follows:

$$\Delta ETR_r^{ID} \equiv \sum_{i \in \mathcal{I}} s_{i,r} \frac{R_{i,2015}^{ID} - R_{i,2008}^{ID}}{L_{i,2007}^{ID}} \quad (2)$$

⁹ We focus on wage employees as those are the workers more directly affected by firms' adoption of robots. This focus is also relevant from a policy perspective, as wage employment typically provides a more reliable and higher incomes than self-employment in a developing country like Indonesia.

¹⁰ Indonesia's island geography and often underdeveloped transportation infrastructures make the hypothesis of limited mobility likely to hold.

An obvious problem with using ΔETR_r^{ID} is its endogeneity to local employment changes, as unobserved industry-level shocks may be correlated with both robot adoption and changes in labor market outcomes. For instance, if investment in robots is driven by the need of keeping up with demand in response to a positive shock, the estimated impact of ΔETR_r^{ID} on employment would be upwardly biased.

As in other studies, we address this issue by developing a measure of *automation possibilities* to instrument *actual automation adoption* (Acemoglu and Restrepo, 2020; Dauth et al., 2019). We focus on industry-specific robot penetration in countries that are ahead of Indonesia in terms of robot adoption. OECD countries appear the most suitable reference group for a middle-income country as Indonesia. We match IFR data with 3-digit industry employment figures from the OECD Structural Analysis Database (STAN). For each industry-year pair, we compute the number of imported robots per thousand workers, averaged across OECD countries. Then, the measure of automation possibilities is computed as in (2), but replacing Indonesian penetration with the industry-specific average in OECD countries:

$$\Delta ETR_r^O \equiv \sum_{i \in r} s_{i,r} \frac{R_{i,2015}^{OECD} - R_{i,2008}^{OECD}}{L_{i,2007}^{OECD}}$$

If robots adoption were driven by exogenous improvements in technology, e.g. a reduction in robot prices, then we should observe a positive correlation between adoption in Indonesia and in other countries.¹¹ Online Appendix Figure A1 confirms a strong positive correlation between 2008–2015 changes in robot penetration in Indonesia and in OECD countries.

This pattern is consistent with the hypothesis that Indonesia's automation across industries is driven by global shocks, rather than specific domestic conditions. Our identification is based on the assumption that such shocks to automation possibilities affect domestic trends only through their impact on actual robot use in Indonesia. In Section 2.4 we test the robustness of this assumption to a variety of checks, including testing for the exogeneity of the shift-share instrument and the potential impact of non-robot technologies.

Our baseline results are derived from 2SLS estimates using the following system of equations¹²:

$$\begin{cases} \Delta ETR_r^{ID} = b_0 + b_1 \Delta ETR_r^O + \alpha \Delta D_r + u_p + \epsilon_r \\ \Delta E_r = \beta_0 + \beta_1 \Delta ETR_r^{ID} + \gamma \Delta D_r + u_p + \epsilon_r \end{cases} \quad (3)$$

where ΔETR_r^O is the instrument based on OECD penetration and ΔE_r is the change in employment between 2008 and 2015 in regency r (in logs).¹³

In line with the rest of the literature on automation and local labor markets, ΔETR_r^{ID} takes the form of a shift-share variable.¹⁴ The identification of the parameter of interest, β_1 , arises from variation in two components: the “shift” in industry robot penetration and the “share”, i.e. differences in local exposure to industries hosting robots.

Our identifying strategy relies on the assumption that no other unobservable factors are correlated with both automation opportunities and changes in employment at the regency level. The first difference transformation of Eq. (3) absorbs all such possible factors whose effect is time-invariant, e.g. geography and factor endowment. However, provinces have labor market powers in Indonesia, including minimum

wage and the enforcement of labor regulation. Differences in regulation might induce differential trends, in turn affecting initial industry composition and employment changes. For this reason, we include province fixed effects, u_p , in (3).

An important concern is that *global* industry shocks could result in a violation of the exclusion restriction. For instance, a positive global demand shock to an industry could increase employment in an Indonesian regency hosting that industry. It could also enhance robot adoption in that industry (including in the OECD), aimed at keeping up with the higher demand. To mitigate such concerns, we construct a regency-level demand shifter, ΔD_r , and include it in (3). The shifter aggregates global exports by industry (excluding Indonesian industries) at the regency-level, using the shares in (1).¹⁵

Another important concern is that domestic penetration of robots might be affected by foreign demand for intermediate inputs. Results from Faber (2020), Artuc et al. (2019), Kugler et al. (2020), and Stapleton and Webb (2020) suggest that robot adoption in advanced economies boosts demand for inputs produced in developing countries. Therefore, our instrument might be correlated with Indonesian robot penetration through increased input demand, thus violating the exclusion restrictions. For this reason, in the vector of controls we also include a measure of industry-specific changes in average OECD imports from Indonesia. As for the demand shifter, the measure is constructed by aggregating these OECD sectoral imports at the regency-level on the basis of industries' employment shares. The bilateral sectoral import data are taken from the OECD Trade in Value Added (TiVA) database.

In Section 2.4, we also test the robustness of the results to controlling for a variety of other baseline covariates, which might be related to time-varying shocks correlated to both automation possibilities and changes in employment.

We cluster standard errors at the regency-level. We also experiment below with alternative clustering and calculate shift-share standard errors, following the methodology of Adao et al. (2019).¹⁶ We weigh the estimate by regencies' population in the base year so as to obtain nationally representative estimates.

2.3. Local labor market results

Column (1) of Table 1 presents the first stage estimates, which confirm the strength of the instrument. Regency-level exposure based on OECD robot penetration is a strong predictor of Indonesian exposure. The coefficient is significant at the 1 percent level and the regression explaining over half of the overall variation in exposure.

The dependent variables in columns (2)–(5) are regencies' employment changes. Column (2) presents the estimated β_1 coefficient for total employment, which is positive, although not statistically significant. Our first result is that industrial robots do not lower total employment *at the local level*. The coefficient is positive although it is estimated imprecisely so it is not significant. This contrasts with the local labor market findings of Acemoglu and Restrepo (2020), who find negative employment effects of robots in US local labor markets. On the other hand, our result is consistent with the evidence in Dauth et al. (2019) for Germany, who find that the significant displacement effects in manufacturing were offset by faster job creation in services. Therefore,

¹¹ For instance, an enhanced machine-vision technology applicable to painting auto parts, should spur robot adoption in the motor vehicles industry worldwide.

¹² We also show that our results are very similar if we estimate OLS reduced forms.

¹³ All employment changes in our dataset are positive and so we do not face issues of zero or negative values.

¹⁴ See Section 2.4 for a discussion of the conditions for consistent identification with a shift-share variable in our context and a battery of diagnostic tests.

¹⁵ Specifically, we calculate $D_r \equiv \sum_{i \in r} s_{i,r} \cdot \Delta EXP_i^{glob}$, where ΔEXP_i^{glob} is the percentage change in total global export excluding Indonesia between 2008 and 2015 in industry i .

¹⁶ Adao et al. (2019) show that shift-share designs, such as those used in this part of the analysis, might lead to residuals that are correlated across regions with similar industry shares and so over-rejection of the null hypothesis of no impact. In our framework, this is important because if there are omitted variables correlated to industry composition and employment changes, these would enter the error term and introduce correlation across regencies.

Table 1
Regency-level results: employment in manufacturing and non-manufacturing sectors.

	(1) Regency ETR	(2) Δ Total employment	(3) Δ Manufacturing employment	(4) Δ Services employment	(5) Δ Agr/Mining employment
Regency ETR (instrument)	0.430*** (0.081)				
Regency ETR		0.051 (0.038)	0.307*** (0.104)	0.021 (0.058)	−0.281** (0.139)
Observations	276	276	276	276	276
R-squared	0.472				
Province FE	Yes	Yes	Yes	Yes	Yes
Regency demand shifter	Yes	Yes	Yes	Yes	Yes
OECD imports	Yes	Yes	Yes	Yes	Yes
First stage F-stat	–	28.36	28.36	28.36	28.36

Notes: this table presents 2SLS estimates of the relationship between regency-level exposure to robots and employment. Indonesian exposure to robots is instrumented with exposure based on OECD industry penetration. The dependent variables are the 2008–2015 differences of log of employment in each regency. OECD imports aggregates sectoral OECD imports from Indonesia by industry using regency-level industry employment shares in the base year. The regency demand shifter aggregates global exports by industry (excluding Indonesia) using regency-level industry employment shares in the base year. Standard errors are clustered at the regency-level. Weights are constructed using 2007 (base year) regency population. The coefficients with *** are significant at the 1% level, with ** are significant at the 5% level, and with * are significant at the 10% level.

in columns (3)–(5) we examine the employment impact of robots in manufacturing, services and agriculture.

Column (3) focuses on manufacturing, which is the robot-adopting sector. The estimates suggest that robots have a positive long-run impact on manufacturing employment, significant at the 1 percent level. The coefficient implies that one additional robot per thousand workers corresponds to 31 percent higher regency-level employment growth over the 2008–15 period. Alternatively, a doubling in the number of robots relatively to the mean (0.16) would generate an increase in manufacturing employment growth of 5 percentage points.

This is the central result of our paper: the adoption of industrial robots in Indonesia has a sizable positive impact on local manufacturing employment. This *local labor market* effect stands in contrast to the negative impacts found by Acemoglu and Restrepo (2020) and Dauth et al. (2019) on local *manufacturing* employment in the United States and Germany, respectively. Giuntella and Wang (2019) and Artuc et al. (2019) provide evidence of similar negative impacts for China and Mexico, respectively. Much of the remainder of the paper will focus on testing the robustness of this important discrepancy and explaining its underpinnings.

Columns (4)–(5) present the results of the impact of robots on other non-manufacturing sectors. The coefficient for services is positive, but small and not statistically significant (column 4). Conversely, the impact of automation on employment in primary sectors (agriculture and mining), is negative and statistically significant (column 5). The results suggest a reallocation of labor towards manufacturing as a result of automation in manufacturing. This accelerates the process of workers moving out of agriculture in particular, as mining represents a small share of workers in primary sectors in Indonesia.

2.4. Threats to the validity of the estimates

We start by checking the robustness of one of the model's key assumptions, i.e. that initial sectoral specialization is not correlated to unobserved regency-level factors with an impact on employment changes. Online Appendix Figure A2 plots the raw correlation between regency-level manufacturing employment changes and base-year regency industry shares. Reassuringly, the two variables appear to be largely uncorrelated in our sample.

Given that the largest changes in robots penetration are concentrated in few industries (Figs. 4 and A1), the identification of β_1 should arise mainly from local differences in exposure to a few common shocks.¹⁷ We follow the approach in Goldsmith-Pinkham et al. (2020)

¹⁷ Borusyak et al. (2018) for an alternative interpretation based on the assumption of quasi-random variation in shocks across many industries.

and compute the so-called “Rotemberg weights”, i.e. relative importance of each industry in determining the overall explanatory power of the shift-share instrument. As in Goldsmith-Pinkham et al. (2020), we find that a small number of industries have a large share of the weight. In our sample Food products and Basic metals together account for over 95 percent of the positive weight of the estimator.¹⁸

We then probe the validity of our identification assumption focusing on such two industries by performing two tests suggested (Goldsmith-Pinkham et al., 2020). Online Appendix Table A1 regresses the regency-level shares of the Food products industry (column (1)) and Basic metals (column (2)) on several base-year covariates that might affect industrial composition as well as changes in employment. In column (3), we repeat the exercise for the ETR instrument as a whole. We do find that some covariates enters the regression significantly and so we re-run the baseline specification including such controls. Reassuringly, however, Online Appendix Table A2 shows that the results are virtually unchanged. Our conclusions are also unchanged if we exclude altogether Food products and Basic metals (Online Appendix Table A3), or Motor vehicles, which experienced the largest increase in robot penetration over the sample (Online Appendix Table A4). Taken together, these robustness checks reassure us against the possibility that the results are driven by specific industries, or that the potential correlation between regency industry shares and changes in employment endanger the validity of the estimates.

We test further the validity of the exclusion restriction by providing evidence that automation possibilities are uncorrelated to other trends affecting Indonesian industries. In particular, we are concerned that the adoption of computers or other non-automation technologies, potentially complementary to robot use, might be driving the positive employment impact. To that end, we estimate industry-level correlations between OECD robot penetration and an index of technological sophistication including information on IT usage, R&D, product and process innovation.¹⁹ The results are presented in Online Appendix Figure A3. In Panel (a), we correlate the change in industry-level OECD penetration between 2000 and 2006 (in logs), to the industry average value of the technological index in 2006.²⁰ The figure shows that there

¹⁸ In their application, Goldsmith-Pinkham et al. (2020) find that only 5 of the available 228 industries explain almost half of the predictive power of the Bartik variable. Therefore, it is not surprising that in our sample with 13 industries, only two explain most of the weight.

¹⁹ The index is based on plant-level data. See Section 4.3 for details on its construction.

²⁰ Ideally, we would have wanted to correlate changes in OECD penetration to changes in computer usage. However, Sakernas labor market data do not provide information on technology usage. Therefore, we exploit plant-level data to calculate industry averages. Since information on technology usage is

is no correlation between the two variables. Panel (b) presents the same relationship for capital intensity, measured as the average capital-labor ratio (in logs). In this case, we find a mild but positive relationship between the two variables. Panel (c) shows instead the correlation between 2008–2015 changes of OECD robot penetration and changes in the capital stock over the same period. Also in this case, the correlation is essentially flat.

In order to further mitigate the concern that our instrument might be correlated to the adoption of non-robot technologies, we check whether the 2008–2015 changes of OECD robot penetration is correlated with changes in capital intensity across regencies. The results are presented in column (1) of Online Appendix Table A5. Reassuringly, we find no evidence of correlation between the two variables. In column (2), we regress the log-change in the aggregate capital stock on the instrument. The coefficient is positive, but still not statistically significant.

Next, we verify that changes in manufacturing employment between 2000 and 2007 do not predict our instrument, nor the regency-level shares of industries with the largest Rotemberg weights, Food products and Basic metals.²¹ This is shown in columns (1)–(3) of Online Appendix Table A6. None of the coefficients is statistically significant, which reassures us about the absence of significant pre-trends. Before turning to the robustness of the labor market estimates of Table 1, we show graphically that regency-level employment changes between 2000 and 2007 were not correlated to employment changes between 2008 and 2015. The results are presented in Online Appendix Figure A4, which shows no sign of correlation between employment growth across the two period.

The labor market results in Table 1 are robust to several alternative specifications. First, we check if our main result depends on whether we estimate the model with OLS or 2SLS. Online Appendix Table A7 presents reduced-form estimates using the OECD-based exposure instrument in an OLS regression. The coefficients are smaller than the 2SLS estimates, but the signs and significance of the coefficients are analogous to those of the baseline.

Second, we estimate a model in stacked-differences by splitting our sample between the years 2008–2009 and 2010–2015, which Fig. 4 shows to exhibit different trends in robot penetration. Online Appendix Table A8 shows that the magnitude of the coefficients is largely unaffected, but they are more precisely estimated than the baseline in long-differences.

Third, automation might be not the only technological force affecting Indonesian employment. In particular, other forms of routine-biased technical change (Autor et al., 2003; David and Dorn, 2013) might be driving our estimates. To test whether that is the case, we rely on data from Frey and Osborne (2017), which provide a list of occupations in manufacturing and non-manufacturing industries at high risk of automation.²² We use the list to calculate the share of Indonesian employment at risk of automation in each regency in the base year.²³

only available for the census year 2006, the most sensible way of presenting the evidence is correlating *past* changes in OECD robot penetration to 2006 values of the index.

²¹ Since Sakernas labor market data are not fully representative before 2007, we pool bi-annual 2000 and 2001 waves to maximize representativeness.

²² Specifically, Frey and Osborne (2017) calculate the probability of computerization based on the description of the tasks performed in each occupation. Then, they deem an occupation at risk of automation if the associated probability is greater than 0.7. The calculation of the probabilities relies on information from the 2010 version of the Dictionary of Occupational Titles (ONET), an online tool developed for the US Department of Labor.

²³ Besides information on employment status, sector and work location, Sakernas also includes information on the occupation of the worker identified according to the Indonesian classification (Klasifikasi Baku Jenis Pekerjaan Indonesia—KBJJ). This is compatible with the International Standard Classification of Occupations (ISCO), which can be mapped to the data from Frey and Osborne (2017).

To proxy for the impact of technology more broadly, we control for the base-year value of the regency-level average of the technological sophistication index, and the log of the capital-labor ratio. The results obtained adding the three variables are presented in Online Appendix Table A9. The effects of automation on aggregate and sectoral employment remain very similar to the baseline specification. Interestingly, we find that a large share of routine employment is associated to negative employment changes in manufacturing, but positive changes in services. This is consistent with a reallocation of routine workers from manufacturing to low-skill services (David and Dorn, 2013). More technologically sophisticated regencies tend to have larger changes in total employment, but this is driven by employment gains in services. We do not find any significant impact of regency-level capital intensity on employment.²⁴

Finally, in Online Appendix Table A10, we experiment with alternative computation of standard errors. In round brackets, we present standard errors clustered at the province-, rather than regency-level. This allows for a correlation of the errors at another administrative level, which holds some policy powers relevant to labor markets. Province-level clustered errors tend to be slightly larger, but the coefficients are still statistically significant. Squared brackets include shift-share standard errors, obtained with the methodology in Adao et al. (2019).²⁵ The coefficients on manufacturing employment and agriculture are still statistically significant with shift-share errors.²⁶

3. Diminishing returns from automation

As discussed above, to the best of our knowledge this is the first empirical analysis finding a positive local labor market impact of robots on manufacturing employment. What can explain the difference with the empirical results from advanced and emerging economies?

To answer this question, it is useful to recall some suggestive evidence of diminishing productivity returns to robots unveiled in Graetz and Michaels (2018). Specifically, they show that larger increases in robot density across 17 advanced economies translated into increasingly small gains in productivity, pointing towards diminishing marginal gains from robot adoption. That evidence is based on a sample of high-income countries in 1993–2017, when their robot density was considerably higher than in Indonesia during our period of analysis.

In fact, robot penetration in Indonesian manufacturing was much lower than in all other economies for which labor market evidence on robots exists. Fig. 5 illustrates the extent of the discrepancy. For each of these countries, the figure plots the number of robots per thousand manufacturing workers in the first (if available to us) and last years of the respective analysis.²⁷ Take the last years of the analyses first, as these are close in range for most countries (2014–2016 except the US, which refers to 2007). Indonesia exhibits the lowest robot penetration among the sample of countries by a factor ranging between 9 (*vis-à-vis* Mexico in 2015) and 99 (*vis-à-vis* Germany in 2014). Similar differences apply also when taking the first years of the analyses. Indonesian

²⁴ This is not surprising, as Online Appendix Figure A3 shows that such variables are at best mildly correlated with robot penetration at the industry-level.

²⁵ Specifically, we follow Dauth et al. (2019) and impose the null hypothesis $H_0 : \beta_1 = 0$. Adao et al. (2019) show that the applying such adjustment improves the finite-sample properties of the confidence intervals when the number of industries is relatively small. The computation is performed with the user-written STATA command *reg.ss* (Adão et al., 2020).

²⁶ However, to maximize comparability with closely related studies (e.g. Acemoglu and Restrepo, 2020; Giuntella and Wang, 2019; Chen et al., 2021; Artuc et al., 2019; Koch et al., 2021), we use conventional error clustering in the baseline specification.

²⁷ The data for robots comes from IFR and for manufacturing employment from the ILO and — for China — from UNIDO. For the missing years, we input the manufacturing employment values on the basis of the share of industry in employment and of total employment in those years (again from the ILO).

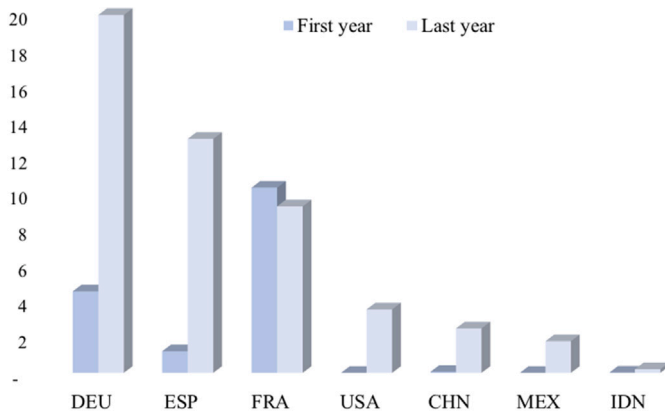


Fig. 5. Robot penetration in Indonesian vs other countries with evidence on the impact of automation (robots per thousand manufacturing workers).

Notes: this figure shows the number of robots per thousand manufacturing workers in the first (if available to us) and last years of the respective countries' analysis. These are: 1994–2014 (DEU); 1990–2015 (ESP); 2010–15 (FRA); 1993–2007 (USA); 2006–16 (CHN); 2000–15 (MEX); 2008–15 (IDN).
Sources: IFR; ILO; UNIDO.

robot density relatively to the other countries varies between a factor of 4 (vis-à-vis China in 2006) and 158 times smaller (vis-à-vis Germany in 1994).

To the extent that diminishing productivity returns to automation are at play, this could translate in increases in labor demand at very low densities of robots, as in the case of Indonesia. This could also explain the much less negative employment impact of automation in Mexico (Artuc et al., 2019) relative to the US (Acemoglu and Restrepo, 2020). We illustrate this idea using a task-based model along the lines of Acemoglu and Restrepo (2018).

3.1. A simple task-based model

Consider an economy in which the unique final good is produced by a representative firm combining a continuum of varieties with total mass equal to 1. The parameter $\sigma > 1$ governs the constant elasticity of substitution across varieties.

Each differentiated variety is produced by a single intermediate good-producing firm f , which combines a unit measure of tasks. Let $x_f(z)$ represents the production services of task z . The output of firm f is given by

$$y_f = \exp\left(\int_0^1 \ln x_f(z) dz\right)$$

Since all firms are identical and face the same problem, we suppress f subscripts and focus on a representative firm.

Task services are produced using machines and human labor, which are perfectly substitutable. Markets for both inputs are perfectly competitive. However, humans are more productive at performing certain tasks. Let $-\infty < \tilde{\gamma}(z) < \infty$ be the log-productivity of labor relative to machines in performing task z , which is the same across firms.²⁸ Without loss of generality, we assume that $\tilde{\gamma}(z)$ is increasing in z , so that higher-ordered tasks are harder to automate.²⁹ Let κ denote the given extent of firm-level automation, such that tasks $z \in [0, \kappa]$ are performed

²⁸ This assumption is more likely to hold for firms operating in industries characterized by similar production processes. To account for this explicitly, one possibility is assuming that the final good is industry-specific and final consumers maximize utility over a continuum of final goods. However, this would complicate the model without altering our conclusions.

²⁹ For instance, these could be non-routine tasks requiring creativity (Autor et al., 2003).

by machines and tasks $z \in (\kappa, 1]$ by labor. In partial equilibrium, firms take factor prices are given. Let n represent labor demand.

This model is a simplified version of the task-based model in Acemoglu and Restrepo (2018).³⁰ Online Appendix E shows that the employment effect of automation is increasing in the productivity effect $\left(\frac{\partial \ln y}{\partial \kappa}\right)$ and decreasing in the displacement effect $\left(\frac{1}{1-\kappa}\right)$:

$$\frac{\partial \ln n}{\partial \kappa} = \frac{\sigma - 1}{\sigma} \frac{\partial \ln y}{\partial \kappa} - \frac{1}{1 - \kappa} \quad (4)$$

Let ω be the log price of labor relative to capital. Online Appendix E shows that:

$$\frac{\partial \ln y}{\partial \kappa} = \sigma(\omega - \tilde{\gamma}(\kappa))$$

The productivity effect of automation for firm f is increasing in ω , because substituting labor with machines reduces marginal costs more when labor is relatively expensive. However, the productivity effect is lower for firms in contexts where a larger number of tasks is already automated (i.e. κ is larger):

$$\frac{\partial^2 \ln y}{\partial \kappa^2} = -\sigma \tilde{\gamma}'(\kappa) < 0$$

This arises from the fact that the $\tilde{\gamma}(\cdot)$ is increasing, which reflects the comparative advantage of humans in performing non-routine, complex tasks. In a context like Indonesia during our period of analysis, where only a relatively small share of tasks is automated, there are more opportunities for productivity gains. On the contrary, in a context where producers have exploited most available opportunities to automate, using machines to perform the marginal task would deliver limited gains. That is because of the strong comparative advantage of humans in performing complex tasks that are not yet automated.³¹

The displacement effect, on the other hand, is always negative and monotonically decreasing in κ . This is due to the fact that as the range of tasks performed by humans shrinks due to automation, the marginal product of labor decreases and workers become redundant. The previous conditions imply that the employment effect is also decreasing in the initial level of automation:

$$\frac{\partial^2 \ln n}{\partial \kappa^2} = -(\sigma - 1) \tilde{\gamma}'(\kappa) - \frac{1}{(1 - \kappa)^2} < 0$$

Thus, for countries where automation is already widespread, increased automation is more likely to decrease employment not only because the productivity effect is lower, but also because the displacement effect is stronger. This happens because if a wide range of tasks are already automated, the labor displaced by the additional automation has a low marginal productivity given the marginal decreasing returns to labor with respect to the range of activities.

The task-based model thus suggests that that the positive employment impact of automation presented in Section 2 might reflect a large productivity effect offsetting the labor displacement effect. This is more likely to happen in a country where firms tend to be less automatized, as in Indonesia.

4. Diminishing returns in plant-level data

This section tests the predictions of the task-based model using Indonesian plant-level data. It describes in turn the data, methods and results of the empirical analysis.

³⁰ Specifically, the more general version in Acemoglu and Restrepo (2018) has: (i) intermediate good-producers combining a varying measure of tasks, which is itself an endogenous variable rather than being a fixed unity mass, and (ii) task services are combined through a constant elasticity production function rather than a Cobb–Douglas.

³¹ Acemoglu and Restrepo (2019) refers to technologies conferring only limited productivity gains as “so-so technologies”.

4.1. Plant-level data

The main data source for the plant-level analysis is the Indonesian survey of manufacturing plants with at least 20 employees (Statistik Industri, SI henceforth). This is an extensive panel of plants, which becomes a census at ten-year intervals and covers over 90 percent of the relevant population in the inter-census years. It includes detailed data on production and sales, including quantities produced by products, on employees, costs and assets, and specifically in machinery and equipment (which includes industrial robots). Online Appendix C.2 provides more detailed information about the plant-level data.

We match SI and IFR data by constructing industry crosswalks, which are presented in Online Appendix C.3.³²

Our final plant level (unbalanced) panel includes 15,351 plants operating in 13 3-digit manufacturing industries between 2008 and 2015, for a total of 65,573 observations.³³ Table A.2 presents summary statistics for the variables used in the plant-level analysis.

4.1.1. Productivity measurement

To measure productivity, we estimate translog production functions using plant-level data. We exploit information on quantities and values of each product produced by plants, including multi-product ones, and construct plant-specific price indexes, as in Eslava et al. (2004) (see Online Appendix C.6). This is an advantage over contributions employing industry-level price deflators, which essentially assume that all plants within an industry face similar inputs' cost and charge the same price.³⁴ To account for differences in production technologies across industries, we estimate production function parameters separately for each 3-digit industry.

A key challenge in the measurement of productivity relates to the endogeneity of the firm's optimal choice of inputs, which we address by employing the control function approach of Akerberg et al. (2015). As in De Loecker and Warzynski (2012), we allow plants' exposure to robots to affect the expected value of future productivity.³⁵ This approach allows us to obtain a measure of quantity-total factor productivity (TFPQ), which reflects changes in pure technical efficiency, rather than in revenues. Online Appendix G describes the details of the estimation procedure.³⁶

4.2. Measuring plant-level exposure to robots

Unlike some other studies in high-income countries (e.g. Acemoglu et al., 2020b and Koch et al., 2019), we do not observe the use of robots by plants. Instead, we match data on Indonesian imports of industrial

³² Industry classifications in both SI and IFR data are roughly equivalent to ISIC Rev. 4. However, in some cases SI industries are more granular than IFR. Thus, we group together some SI industries to ensure maximum compatibility across the two datasets.

³³ In principle, we could run the plant level analysis starting in 2007. However, representative data by regency and industry, which are discussed in Section 2.2, are only available from 2008. Therefore, to be consistent we start the plant-level analysis in 2008. Plant level results for the period 2007–2015 are very similar, which is not surprising given that robot adoption started to take off in 2009 (see Fig. 3).

³⁴ Foster et al. (2008) discuss the bias arising when using plant revenue deflated by industry deflators. De Loecker et al. (2016) extend the analysis in the context of unobserved variation in input prices.

³⁵ Similarly, De Loecker et al. (2016) study the impact of trade reforms and include export dummies and import tariffs; De Loecker (2007) includes export quotas; Doraszelski and Jaumandreu (2013) include R&D expenditure, and Konings and Vanormelingen (2015) include measures of workforce training.

³⁶ We exclude outliers in the top and bottom three percent of the productivity distribution, which correspond to implausibly extreme values of the productivity measures.

Table 2

Share of employed workers in occupations at high risk of automation, by educational attainment.

Sources: Sakernas; Graetz and Michaels (2018).

Educational attainments:	Primary	Secondary	Tertiary
Share of employment			
At high risk of automation	.39	.59	.02

Notes: this table reports the share of employment in production occupations at high risk of automation, by the educational attainments of Indonesian workers in 2006. Occupations at risk of automation are identified using the methodology of Graetz and Michaels (2018). Primary education includes up to completed primary school. Secondary education includes junior and senior high-school. Tertiary education includes education levels from diplomas to PhD.

robots by industry with plants' observable characteristics to build a measure of plant-level exposure to robots.

The key assumption underlying our proxy of plant exposure is that certain occupations are more likely to be automated (Autor et al., 2003). As we do not observe the tasks performed by workers in each plant, our baseline measure of exposure proxies employment amenable to automation with the educational level of each plants' workforce, which we observe in 2006 when the manufacturing plants' survey becomes a census.

Specifically, we identify the production occupations most exposed to being automated using the definition of occupations' "replaceability" from Graetz and Michaels (2018). Then, using Indonesian labor force survey data (described in Section 2.1), we compute the distribution of workers in occupations at high risk of automation by their educational attainments in 2006, the first year such information is available. Table 2 presents the result and shows that workers in occupations at risk of automation in Indonesia are typically secondary education workers.³⁷

The finding that occupations at risk of automation are dominated by secondary educated workers is consistent with various pieces of evidence. First, the literature on employment polarization suggests that automation technologies tend to replace occupations with an intermediate level of skills.³⁸ While we do not observe the skill level of plants' workforce directly, secondary education is both the median and the mean level of education of workers in manufacturing plants in 2006.

Second, an analysis of the characteristics of production occupations compiled by the World Bank for Indonesia, Thailand and Malaysia, suggests that secondary education is the typical educational attainment for all of the occupations at greater risk of automation across countries with similar economic characteristics (see Online Appendix D for details on the Occupation Profiles).³⁹

Third, industries with an initial large share of secondary education workers have adopted relatively more robots in subsequent years (Online Appendix Figure A5). In addition, Online Appendix Figure A6 shows that industry averages of investment in machinery and equipment from SI, which includes investment in robots, is positively correlated to industry imports of robots from IFR data.

Formally, we define exposure to robots (ETR) for plant f as:

$$ETR_{f,t} \equiv \frac{R_{i,t}}{L_{i,t_0}} \times secondary_{f,t_0} \quad (5)$$

³⁷ Similar results are obtained using the probabilities of computerization calculated by Frey and Osborne (2017) (see Cali and Presidente, 2021). However, while Frey and Osborne (2017) consider production and non-production occupations, the notion of replaceability in Graetz and Michaels (2018) is more closely related to industrial automation. Given the focus of the paper on robots, using the list of occupations from Graetz and Michaels (2018) seems more appropriate in our framework.

³⁸ E.g. Goos et al. (2014), David and Dorn (2013), Goos et al. (2009) and Goos and Manning (2007).

³⁹ We thank Mauro Testaverde for sharing these unpublished profiles with us.

where $secondary_{f,t_0}$ is the plant-level share of workers with secondary educational attainments in the base year. Plants that at the beginning of the sample had a large share of secondary education workers are expected to be more exposed to robots as industry penetration grow. On the contrary, ETR is zero if either a plant operates in an industry that does not import robots, or if $secondary_{f,t_0} = 0$ and so the plant does not have opportunities for automation.

To probe further the validity of our assumption that secondary education proxies for routine task-intensity in Indonesian manufacturing, we regress plant-level investment in machinery and equipment in every year on plant-specific ETR controlling for various plant-, year- and industry-year specific controls.⁴⁰ The result is presented column (1) of Table A.3. Reassuringly, the ETR coefficient is positive and statistically significant. It suggests that one additional robot per thousand workers in a given industry is associated with a nine percent increase in machinery and equipment investment by plants one standard deviation above the average share of secondary education workers in that industry.

In columns (2) and (3), we report the results of two placebo tests. The first shows that investment in machinery and equipment is *inversely* related to the share of primary education workers — typically performing non-routine manual tasks that are hard to automate. The second shows that these investments are not correlated with the share of workers with tertiary education, consistently with the idea that they tend to perform complex non-routine cognitive tasks (Autor et al., 2003).

We take this evidence to corroborate the validity of our approach, which identifies exposure to robots using secondary education workers. However, one might still be concerned that our exposure measure may miss the impact on primary educated workers, as almost forty percent of these workers (and two percent of tertiary educated workers) are at risk of automation.⁴¹ Therefore, in column (4) we employ an exposure measure which replaces the plant-level secondary employment share with the plant-level weighted employment shares where the education-specific weights are taken from Table 2. The resulting coefficient is very similar to the baseline specification in column (1), which suggests that the entire effect is driven by secondary education workers. Nonetheless, we perform robustness checks in Section 4.5 with such an alternative combined exposure variable.

4.3. Plant-level analysis: Econometric specification

We use ETR to identify the impact of automation on plant-level outcomes by estimating the following linear model⁴²:

$$Y_{f,i,t} = \gamma_0 + \gamma_1 ETR_{f,t} + \Gamma X_{f,t} + \eta_f + u_{i,t} + \epsilon_{f,t} \quad (6)$$

where $Y_{f,i,t}$ is an outcome of plant f in industry i at time t ; $X_{f,t}$ is a vector of controls, to be discussed further below; η_f are plant

⁴⁰ Specifically we employ (6), which is described in the next section, replacing the dependent variable with investment in machinery. Given the lumpiness of investment at the plant level, for the dependent variable we employ the log-transformation $\ln[x + (x^2 + 1)^{-5}]$. We obtain qualitatively identical results using a dummy variable taking value 1 if, in a given year, a plant has positive investment.

⁴¹ For instance, Graetz and Michaels (2018) find a small but significant negative impact of robots on low-skill employment and positive on high-skill workers.

⁴² The plant-level approach is different from the local labor market analysis in that it exploits yearly variation in robot penetration. We argue that the yearly analysis is appropriate for two reasons. First, running the plant level specifications in long difference would reduce the sample size by almost two-thirds relatively to the yearly regression model, as it would only focus on surviving plants throughout the entire period. Labor force survey data are not subject to this problem, which allows us to estimate models in long differences. Second, dropping plants that were in the sample in 2006 but exited before 2015, might introduce selection bias.

fixed effects, and $u_{i,t}$ are industry-year effects defined using the same classification as IFR robot import data.

Plant fixed effects absorb the confounding effect of any plant-level unobserved time-invariant characteristic. This includes factors that might be related to plants having a large base-year share of secondary education and their outcomes. We analyze this aspect in detail further below.

Industry-year fixed effects mitigate the concern that exposure to robots in an industry may be endogenous to the outcomes of plants operating in that industry. They capture any industry-specific, time varying shock that may be related to both the outcome variable and robot penetration, such as changes in international trade patterns, or demand and supply shocks, including the sourcing of intermediate inputs from advanced economies documented in Artuc et al. (2019) among others. At the same time, $u_{i,t}$ absorbs the variation of industry-specific robot penetration, $\frac{R_{i,t}}{L_{i,t_0}}$, which therefore cannot be estimated.

Estimates of γ_1 in (6) quantify the *net* impact of one additional robot in exposed plants (i.e. high levels of $secondary_{f,t_0}$), relatively to less exposed ones (i.e. low levels of $secondary_{f,t_0}$) within the same industry. A possible threat to the validity of the estimate is externalities from treated to control plants (Miguel and Kremer, 2004). For instance, an increase in robot penetration in industry X raising productivity of exposed plants in X could also benefit less exposed plants in the same industry through an increase in demand for inputs. The industry classification used by IFR data is sufficiently broad that most inputs are typically sourced within the same industry. Hence this increase in demand could translate into higher sales and employment for upstream producers within the same industry regardless if they are automated.⁴³ This would result in downward-biased estimates of γ_1 . In order to obtain a clean control group, we exploit 9-digit-level information on inputs purchased by each plant and obtain a detailed input-output table (see Online Appendix C.5). We then use this table to compute downstream exposure to robots in each 5-digit industry and include it in $X_{f,t}$.⁴⁴ The details of the construction of the downstream exposure variables are presented in Online Appendix F.

Similarly to Section 2.2, an important concern with Eq. (6) is that the adoption of other non-automation technologies might bias our estimates if adoption were correlated to the share of secondary educated workers. Specifically, skill-biased and routine-biased technical changes might affect differentially firms on the basis of their skill composition (Bresnahan et al., 2002). This would result in unobserved trends in technology adoption that might bias the estimates. We address such concerns in various ways.

First, we absorb skill-biased technical change through two sets of controls: a plant-level measure of innovation and the plant-specific share of high-skilled employment (both measured in the base-year). We then interact each control with year effects. The measure of innovation intensity is a composite index of R&D units and product or process innovation, exploiting plant-level information available in 2006 (census year).⁴⁵ High-skilled employees are defined as having high school diploma or above.

Second, we control for other forms of automation that are not strictly related to industrial robots, by constructing an index which takes the value 1 if the plant owned a computer and had access to the

⁴³ Cali and Presidente (2021) document significant employment spillovers to non-exposed plants from downstream automation in Indonesia.

⁴⁴ Similarly to Section 2.2, the downstream indicators are based on OECD average robot penetration and they are interpreted in terms of automation possibilities. This allows us to avoid concerns of endogeneity with plant-level outcomes.

⁴⁵ The index ranges between 0 and 1. It is computed by taking the average of three dummies, each equals 1 if a plant had any R&D unit, performed product innovation, or process innovation in 2006.

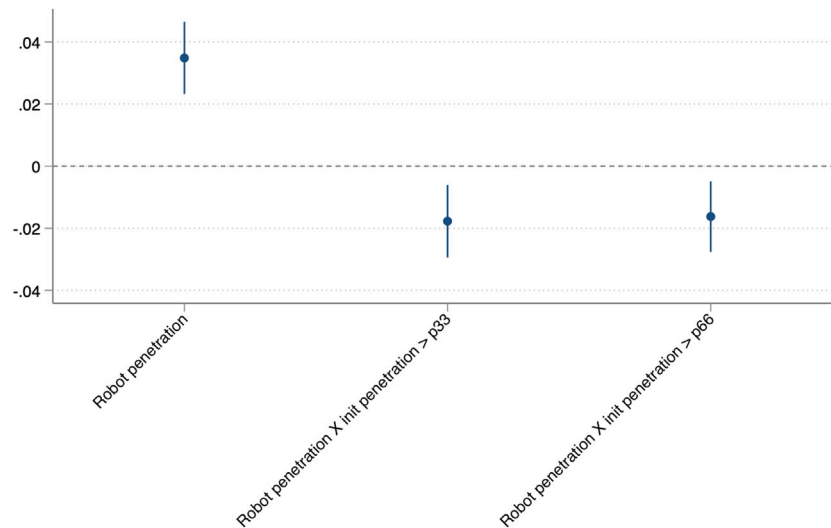


Fig. 6. Decreasing returns from automation: employment impact of robot penetration by tercile of initial robot penetration.

Notes: this figure shows point estimates and 95% confidence intervals for the relationship between plant-level log-employment and the number of robots per thousand employees (robot penetration), by tercile of initial exposure. To calculate initial exposure, for each industry, we allocate plants into terciles based on the baseline share of secondary education workers in their workforce. We consider plants with the highest shares to be the least automated at the beginning of the period. The interaction coefficients represent the marginal effects of robot penetration for plants above each terciles of the distribution of initial penetration.

Sources: Authors' calculations based on IFR and SI.

internet in the base year and 0 otherwise. Interacting this index with year effects should capture the possible impact of a key form of routine-biased technical change on plants' employment (and productivity). We include all such variables in $X_{f,t}$.

Third, we perform robustness tests by interacting secondary educated non-production workers with year effects to control more directly for the impact of technology on non-manual routine tasks.⁴⁶ To the extent that innovation activity, skill-biased and routine-biased technical change are correlated to the propensity of plants to adopting technologies other than robots, this approach should help in purging estimates of γ_1 from their impact.

Given that robot penetration in (5) varies at the 3-digit industry- and year-level, we would ideally employ industry-year two-way clustered errors. We have 13 industries in our data, therefore for the baseline estimates we prefer to cluster errors at the industry-year level to increase the number of clusters, which are thus 117. Section 4.5 presents robustness tests with alternative clustering (including two-way clustering).

To ensure consistency, in the plant-level analysis we drop observations with missing information on either productivity or employment, as these are linked by a causal relationship in the model of Section 3.1. Not implementing this procedure, however, does not alter our conclusions.

Finally, we normalize $secondary_{f,t_0}$ so that it has sample average value equal to zero and unitary standard deviation.⁴⁷ This allows us to interpret the estimated coefficient as the impact of one additional robot on a plant with high opportunities for automation.

4.4. Plant-level results

Table 3 displays the main results of running Eq. (6) on our plant data. The *ETR* coefficient in column (1) is positive and statistically significant. It implies that on average, one additional robot per thousand workers increases the plant's employment by 1 percent. This average plant-level estimate is consistent with the positive employment impact documented in the local labor market analysis in Section 2.2.

4.4.1. Productivity effects

The model in Section 3.1 posits that the positive employment effect of automation is a product of the positive productivity effect offsetting the negative displacement effect. This section tests the validity of this result by examining the effect of robots on plants' productivity.

We measure productivity by using the TFPQ measure as the dependent variable in specification (6). The results — presented in Table A.4 — confirm the large productivity impact of automation. One additional robot per thousand workers increases TFPQ by 7 percent on average (column 1).

Column (2) check the robustness of the results to using an alternative productivity variable: real marginal costs.⁴⁸ The results are consistent. The coefficient in column (2) implies that one additional robot per thousand workers lower marginal costs by 10 percent on average.

4.4.2. Diminishing returns from automation

To test our hypothesis of diminishing returns from automation, we estimate the impact of robot penetration semi-parametrically, exploiting cross-plant differences in initial robot penetration. To estimate plants' initial penetration, we follow the same logic used (and validated) so far, i.e. robots mainly substitute secondary education workers. Accordingly, for each industry, we allocate plants into terciles based on the baseline share of secondary education workers in their workforce. We deem plants with the highest shares to be those with the least robot penetration at the beginning of the period. We then estimate for each tercile the employment effects of time-varying robot penetration by industry, i.e. $\frac{R_{i,t}}{L_{i,t_0}}$.

Fig. 6 shows that the employment impact of robot penetration is positive and significant for plants in the bottom tercile of initial penetration, i.e. those with most abundant opportunities for automation. This effect then drops significantly for plants in higher terciles. We address a number of possible threats to the robustness of this result in the following section.

⁴⁶ Data on non-production employment by educational attainment are more coarse as not all plants report it.

⁴⁷ The actual sample average value of $secondary_{f,t_0}$ is 0.616 (see Table A.2).

⁴⁸ After estimating the parameters of the production functions industry by industry, we follow De Loecker and Warzynski (2012) and obtain also a measure of plants' marginal cost, based on the plants' first order conditions and flexible inputs' choice.

Table 3
Exposure to robots and plant-level employment: main results.

	(1) Employment	(2) Employment
ETR	0.010** (0.005)	0.022*** (0.006)
ETR × high-initial penetration		−0.019*** (0.007)
Observations	65,573	65,573
R-squared	0.927	0.927
Plant FE	Yes	Yes
Industry-year FE	Yes	Yes
Other technologies	Yes	Yes
Downstream automation	Yes	Yes

Notes: this table presents OLS estimates of the relationship between plants' exposure to robots (ETR) and log-employment. ETR is defined as industry robot penetration times the plant-level share of secondary education employment in the base year. High-initial exposure plants are plants with base-year ETR larger than the bottom tercile of robot penetration in the cross-section of plants in the base year, which we estimate based on their share of secondary education workers. Other technologies are captured by the base year values interacted with year fixed effects of: (i) an index of plant innovation activities; (ii) an index of computer and Internet usage, and (iii) the plant share of tertiary education workers. Downstream automation captures automation possibilities in downstream industries using 5-digit × 3-digit industry IO tables and average OECD robot penetration interacted with plant-level share of secondary education employment in the base year. Standard errors are clustered at the 3-digit industry-year-level. The coefficients with *** are significant at the 1% level, with ** are significant at the 5% level, and with * are significant at the 10% level.

Since robot penetration varies by industry and year, we cannot include industry-year effects as in the baseline specification (6). Therefore, to assess the quantitative importance of decreasing returns and relate the impact to the baseline specification, we use an interaction term between $ETR_{f,t}$ and a dummy equal to 1 if a plant has initial penetration above the bottom tercile of the sample distribution. The results are presented in column (2) of Table 3, which confirms that the employment elasticity of robot adoption is significantly lower for plants with high-initial penetration. In particular, for high-initial penetration plants, one additional robot per thousand workers is associated with a small and not statistically significant employment elasticity (coeff. = .003, s.e. = .006).⁴⁹ Instead, for the other plants, the estimated elasticity is 2.2 percent (p-value = 0.001)—double the average effect across the entire distribution.⁵⁰ Thus, the estimates are consistent with the implications of the task-based model in Section 3.1.

4.4.3. Discussion

Unlike the results of Section 2, which documented a positive effect of robots on local manufacturing employment, the plant-level evidence is fully consistent with the existing literature from other countries, which documents a positive firm-level employment effect of robots (e.g. Acemoglu et al., 2020b; Koch et al., 2021). It follows that in order to explain the positive local labor market effect in Indonesia, we should find plant-level employment effects that are larger than those documented in advanced economies. The main difficulty in comparing our estimates to existing firm-level studies is that our exposure variable in (5) is expressed in terms of robots per thousand workers, and it is further interacted with the share of secondary education workers. A back-of-the-envelope calculation, however, suggests that at

10%, our average estimated impact of an additional robot on plant-level employment is considerably larger than what is typically found in advanced economies.⁵¹ Specifically it is almost double than the equivalent estimate for Spain (5.8%) (Koch et al., 2021), and 30% higher than in France (7.8%) (Acemoglu et al., 2020b). Thus, the evidence is consistent with the existence of large employment effects of automation in a developing country due to decreasing marginal returns to automation. In turn, this might help explain the positive local labor market effect documented in Section 2 relative to advanced economies.

4.5. Robustness checks

This section examines the robustness of the plant-level results. First, we verify that it is robots, rather than the general firm's investment or the building of new plants, that drive the effect of interest. To do so, we construct an exposure variable similar to ETR but based on the total 2-digit industry real stock values of land, buildings, vehicles—rather than robot imports.⁵² The results are presented in Online Appendix Table A12. None of the exposure variables based on non-robot capital is statistically significant, which is consistent with the idea that the effects on employment are driven only by the robot-specific types of investment.

Second, and related to the previous point—the effects we attribute to automation might be picking up the general economic expansion of the sector and/or the inflows of foreign direct investment (FDI). Those changes in turn might increase both the demand for labor and that for robots.⁵³ To check the robustness of the results to these confounding factors, we exploit two sources of annual data at the 2-digit KBLI industry. The first covers the number and annual value of FDI projects to Indonesia and it is compiled from Indonesia's investment Coordinating Board (Badan Koordinasi Penanaman Modal, BKPM). The second is the total real value added from the United Nations Industrial Development Organization (UNIDO). Using these data, we construct the equivalent of our original exposure variable but replacing the industry-specific robot numbers with industry-specific numbers and value of FDI projects and total value added respectively. The results are presented in Online Appendix Table A13. In columns (1)–(3) the ETR coefficient remains positive and significant while both FDI-based exposure and the value added-based exposure controls are not statistically significant.⁵⁴ These tests reassure us that robots do not act as an observable proxy for the growth of the sector or for FDI inflows.

A more general concern with the plant-level estimation is that since automation is concentrated in a few industries (Fig. 4), there might be too little variation in robot penetration to estimate the parameters of (6). However, this is unlikely to be a problem in our framework, because the identification is mainly based on *within-industry* differences in exposure to robots across plants. Therefore, our identification is valid as long as there is some variation in industry penetration. Moreover, despite the concentration in industry penetration, exposure to robots

⁵¹ Specifically, in our data the average plant employs hundred workers. We assume that adopting plants have a share of secondary education workers one standard deviation above the mean. Thus, we can express the estimated coefficient in column (1) of Table 3 in terms of the plant-level employment impact of adopting one robot by multiplying $0.01 \times 1000/100 = 0.1$.

⁵² Specifically, we aggregate the nominal value of capital stock by asset type provided in the SI data for each 2-digit industry and deflate them with the asset-specific deflator. Then, we divide the quantities by the base-year number of workers in that industry, exactly as we do for ETR. We exclude "machinery" from capital assets in the construction of the index, as it includes industrial robots, as discussed in Section 4.2.

⁵³ FDI inflows in the Indonesian manufacturing sector increased significantly during the period of analysis, due to large greenfield investments in metals, chemicals, motor vehicles, and the food industry (OECD, 2020).

⁵⁴ The number of observations in column (3) is lower due to missing data from the UNIDO dataset.

⁴⁹ This impact corresponds to the sum of the coefficient in column (2).

⁵⁰ Online Appendix Table A11 shows that the baseline results in Table 3 are still valid if we use two-way clustering of the standard errors by 3-digit industry and year. Since the number of 3-digit industries and years is limited, the number of clusters available is low. Hence, we report 90 percent confidence intervals and p-values obtained by implementing a wild bootstrapping procedure with 9999 repetitions, which still delivers significant coefficients.

Table A.1
Summary statistics: local labor market analysis.

	(1) N	(2) Mean	(3) sd	(4) Min	(5) Max
Regency Δ ETR	284	0.165	0.316	-0.278	2.243
Regency Δ ETR (instrument)	284	0.504	0.524	-3.162	1.990
Change in total employment (log)	284	0.122	0.128	-0.178	0.615
Change in manufacturing employment (log)	284	0.121	0.376	-2.839	1.325
Change in services employment (log)	284	0.276	0.210	-0.176	1.373
Change in agriculture and mining employment (log)	284	-0.0920	0.273	-1.400	0.869
Change in RoW export by regency (log)	276	0.0771	0.380	-2.913	0.829
Population in 2007	281	793,849	704,630	29,682	5.756e+06
Share of workers with tertiary education in 2007	281	0.0482	0.0324	0.00898	0.206
Share of workers with no education in 2007	281	0.113	0.0459	0.0178	0.270
Natural resources share of output in 2007	281	0.495	0.233	0.0392	0.983
Share of employment at risk of computerization	284	0.0138	0.0110	0	0.0856
Change in real capital stock (log)	277	1.188	0.475	-0.594	2.536
GDP per capita in 2007 (log)	281	2.934	0.638	1.757	5.518

Notes: this table presents the summary statistics (number of observations, mean, standard deviation, minimum and maximum values) of the variables involved in the local labor market-level analysis. Local labor market markets are identified with regencies.

Table A.2
Summary statistics: plant-level analysis.

	(1) N	(2) Mean	(3) sd	(4) Min	(5) Max
TFPQ (log)	66,877	5.646	20.20	-111.4	207.2
Real expenditure on domestic inputs (log)	65,914	9.708	2.168	-4.034	19.47
Industry number of robots (1000s of workers)	66,877	0.163	0.728	0	14.08
Innovation-intensity (index)	66,877	0.123	0.250	0	1
IT usage (index)	66,877	0.308	0.405	0	1
Share of secondary education workers	66,877	0.616	0.361	0	1
Share of secondary education workers (non-production)	47,002	0.726	0.315	0	1
Share of primary education workers	66,877	0.371	0.368	0	1
Share of tertiary education workers	66,877	0.0131	0.0440	0	1
Downstream exposure to robots (index)	66,877	0.0913	0.369	-0.311	8.996
Downstream exposure to robots (index, OECD)	66,877	1.316	3.685	-29.45	35.75
Real investment in machinery and equipment (log)	66,877	1.761	3.599	0	19.64
Employment (log)	66,877	3.969	0.967	2.197	9.458
Real marginal cost (log)	59,538	2.204	2.090	-19.16	15.02

Notes: this table presents the summary statistics (number of observations, mean, standard deviation, minimum and maximum values) of the variables involved in the plant-level analysis.

exhibits substantial variation even between industries. That is because plants in low-penetration industries might be highly exposed to robots and vice-versa. This can be seen in Online Appendix Figure A7, which shows the industry-level average share of plants with high exposure to robots.⁵⁵ The figure shows that all plants in Motor vehicles, for instance — the industry with the highest robot penetration — can be considered highly exposed. However, most other industries exhibit a substantial share of high-exposure plants, in spite of considerably lower penetration.

Another plausible concern with the estimates of Table 3 is that they may not survive to the use of alternative specifications. We probe the results to various modifications to our baseline specification. We begin by replacing the baseline ETR variable with the weighted version using the share of employment by educational attainments as weights as per Table 2. The results are presented in column (1) of Online Appendix Table A14. The coefficient is very similar to the baseline, confirming that the bulk of the employment impact is driven by secondary education workers.

We address the concern that the impact of robot imports on plants' performance might be driven by omitted variables by adding controls to the regressions, which account for other forms of routine-biased technical change unrelated to robots. Specifically, we include the base-year share of mid-skill workers interacted with year effects, as well as the interaction between year effects and the base-year share of mid-skill

non-production employment, which might be amenable to automation through non-robot technologies such as accounting or logistic software. The results are presented in column (2) of Online Appendix Table A14, which shows that the coefficients are even larger and more precisely identified. However, due to lack of information on non-production employment for many plants, the number of observations is reduced by a third in this specification.

To further address the potential endogeneity of robot imports and plants' performance, we instrument $ETR_{f,t}$ with exposure to robots based on OECD average penetration, i.e. $ETR_{f,t}^{OECD} \equiv \frac{R_{i,t}^{OECD}}{L_{i,t_0}^{OECD}} \times secondary_{f,t_0}$. Column (3) of Online Appendix Table A14 shows that the 2SLS coefficient is positive and larger than the OLS estimate.⁵⁶ Its magnitude implies that one robot per thousand workers increases plant-level employment by roughly 5 percent. The size of the effect is thus similar to the local labor market estimate of Section 2.2.

One important concern of adopting $ETR_{f,t}^{OECD}$ as an instrument is that automation possibilities might be correlated with other factors with a potential impact on Indonesian employment.⁵⁷ We perform a simple test for the validity of the exclusion restriction by exploiting the fact that two industries — Textiles and Wood and paper products — do not experience any robot penetration throughout the period.

⁵⁵ We deem high exposure plants those with exposure higher than the median sample value in at least one year. Results are similar with alternative definitions of high exposure.

⁵⁶ The first stage exhibits an F statistics above 10. The associated first stage coefficient is positive and significant at the 99 percent (coeff. = .372, s.e. = .116).

⁵⁷ For instance, automation possibilities might be correlated to the adoption of computers, which in turn might have an impact on employment.

If $ETR_{f,t}^{OECD}$ captures robot adoption, then plant-level employment should not be affected by the instrument in a reduced form specification for plants in those industries. Column (1) of Online Appendix Table A15 presents the reduced form coefficient obtained regressing plant-level employment on $ETR_{f,t}^{OECD}$, which is positive and significant at the 1 percent level. Column (2) confirms that indeed, for plants in industries with no actual Indonesian penetration, the OECD-based instrument is not statistically significant. These results suggest that $ETR_{f,t}^{OECD}$ has an impact on plant employment only through its impact on automation possibilities.⁵⁸

Finally, we probe the robustness of the evidence in support of decreasing employment and productivity returns. To this end, we re-run the tercile regression of Fig. 6 using the alternative specifications discussed above.

We start with employment in Panel (a) of Figure A8, in which we calculate initial exposure using the combined share aggregating workers of all skills. Panel (b) includes the additional technology controls to capture routine-biased technical change unrelated to robots, and panel (c) computes initial exposure using OECD penetration, as in the IV approach. The figure shows that the estimates are similar to the baseline.

Finally we test whether the finding of decreasing marginal returns is robust to measuring exposure to automation in the current, rather than the base year. Online Appendix Table A16 adds ETR-squared to the baseline specification. The estimated coefficient of the squared term is negative and statistically significant, supporting the presence of decreasing employment returns.

For productivity, Online Appendix Figures A9–A10 present baseline and robustness checks for decreasing marginal productivity returns for TFPQ and marginal costs respectively. The pattern of diminishing returns is broadly in line with that for employment, with stronger productivity effects for plants with low initial exposure.⁵⁹

In sum, the evidence supports the prediction of the model that the positive plant-level employment effect is the product of a relatively large productivity effect offsetting a negative displacement effect, and that the productivity effect declines with the incidence of robots and the exhaustion of automation opportunities.

5. Conclusions

This paper provided novel evidence casting doubts on the hypothesis that automation in developing countries entails a productivity/employment trade-off in the manufacturing sector. The positive impact of robots on local manufacturing employment that we observe in Indonesia is at odds with most of the evidence from advanced and emerging economies alike.

We explain this discrepancy by documenting the presence of diminishing productivity returns to robot adoption in manufacturing. As productivity drives employment demand, we argue that automation is more likely to boost manufacturing employment in a country at the early stages of adoption, such as Indonesia.

In spite of the large productivity gains that we find to be induced by robots, Indonesian manufacturing still exhibits a low level of adoption. Therefore, it is crucial to better understand the potential role of barriers to adoption, and to what extent policy intervention may be appropriate to address them.

Taken at their face value, the results in this paper suggest that automation may be more of an opportunity than a threat even for

⁵⁸ The evidence supporting the validity of the plant-level instrument reinforces also our confidence in the regency-level instrument used in Section 2.2.

⁵⁹ Since the semi-parametric regressions cannot include industry-year effects, the coefficients cannot be directly compared to those in Table A.4. However, the decreasing (increasing) trend for TFPQ (marginal cost) is still visible in the figures.

Table A.3

Correlations between plant level exposure to robots and plant level investment.

	(1) Investment	(2) Investment	(3) Investment	(4) Investment
ETR	0.110** (0.047)			
ETR (primary)		−0.103** (0.040)		
ETR (tertiary)			−0.018 (0.046)	
ETR (combined)				0.109* (0.056)
Observations	65,573	65,573	65,573	65,573
R-squared	0.639	0.639	0.639	0.639
Plant FE	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes
Other technologies	Yes	Yes	Yes	Yes
Downstream spillovers	Yes	Yes	Yes	Yes

Notes: this table presents OLS estimates of the relationship between plants' exposure to robots (ETR) and investment in machinery and equipment. Investment is expressed in log-transformed units: $\ln[x + (x^2 + 1)^{-5}]$. In column (1), ETR is defined as industry robot penetration times the plant-level share of secondary education employment in the base year. In column (2), ETR (primary) is defined as industry robot penetration times the plant-level share of primary education employment in the base year. In column (3), ETR (tertiary) is defined as industry robot penetration times the plant-level share of tertiary education employment in the base year. All education shares are normalized to have zero mean and unitary standard deviation. Other technologies are captured by the base year values interacted with year fixed effects of: (i) an index of plant innovation activities; (ii) an index of computer and Internet usage, and (iii) the plant share of tertiary education workers. Downstream automation captures automation possibilities in downstream industries using 5-digit $\times$ 3-digit industry IO tables and average OECD robot penetration interacted with plant-level share of secondary education employment in the base year. Standard errors are clustered at the 3-digit industry-year-level. The coefficients with *** are significant at the 1% level, with ** are significant at the 5% level, and with * are significant at the 10% level.

the labor markets in economies at early stages of adoption. In that sense, automation could help accelerate the process of economic and wage convergence of developing economies vis-a-vis high-income ones. To the extent that diminishing returns to automation apply across countries, this could also imply increased automation investment flows in countries at early stages of adoption, at least until returns approach convergence.

More generally, the results in this paper suggest that further examination of the impact of automation in developing countries on their own economies is an important topic for future research.

CRedit authorship contribution statement

Massimiliano Cali: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Giorgio Presidente:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Appendix A

See Tables A.1–A.4.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.labeco.2025.102731>.

Data availability

The data that has been used is confidential.

Table A.4
Exposure to robots and plant-level productivity.

	(1) TFP	(2) Marginal cost
ETR	0.070* (0.038)	−0.100*** (0.021)
Observations	62,066	54,683
R-squared	0.994	0.682
Plant FE	Yes	Yes
Industry-year FE	Yes	Yes
Other technologies	Yes	Yes
Downstream automation	Yes	Yes

Notes: this table presents OLS estimates of the relationship between plants' exposure to robots (ETR), TFPQ and real marginal costs. ETR is defined as industry robot penetration times the plant-level share of secondary education employment in the base year. High-initial exposure plants are plants with base-year ETR larger than the 40th percentile of the ETR distribution in the base year. Other technologies are captured by the base year values interacted with year fixed effects of: (i) an index of plant innovation activities; (ii) an index of computer and Internet usage, and (iii) the plant share of tertiary education workers. Downstream automation captures automation possibilities in downstream industries using 5-digit $\times$ 3-digit industry IO tables and average OECD robot penetration interacted with plant-level share of secondary education employment in the base year. Standard errors are clustered at the 3-digit industry-year-level. Outliers in the top and bottom three percent of the productivity distribution are dropped prior to estimation. The coefficients with *** are significant at the 1% level, with ** are significant at the 5% level, and with * are significant at the 10% level.

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